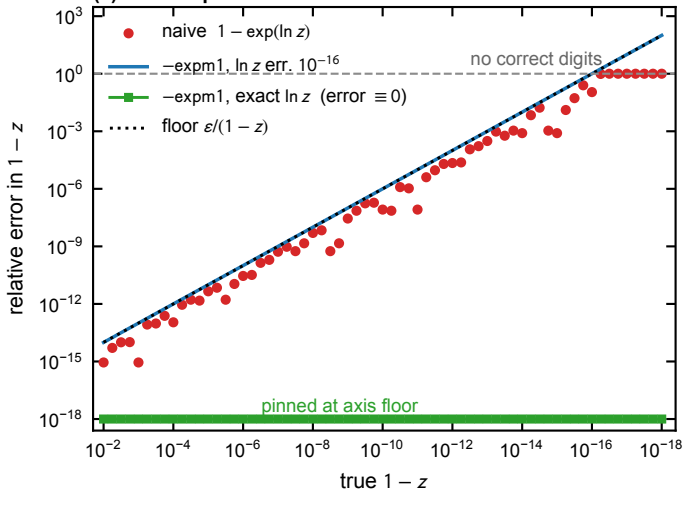
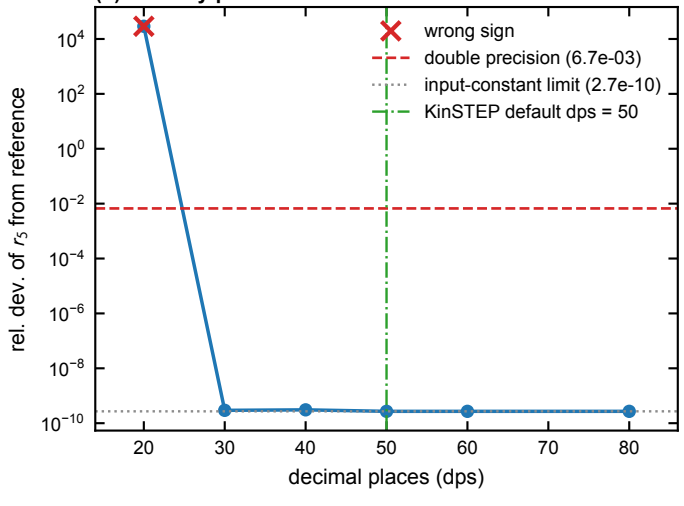

(a) catastrophic cancellation
naive $1 - \exp(\ln z)$
$-\mathrm{expm1}$, $\ln z$ err. 10^{-16}
$-\mathrm{expm1}$, exact $\ln z$ (error $\equiv 0$)
floor $\varepsilon/(1-z)$
no correct digits
pinned at axis floor
relative error in $1 - z$
true $1 - z$
10^{3}
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}


(b) arbitrary-precision escalation
wrong sign
double precision (6.7e-03)
input-constant limit (2.7e-10)
KinSTEP default dps = 50
rel. dev. of r_5 from reference
decimal places (dps)
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
20
30
40
50
60
70
80